

Annexes du rapport de stage (SPE)

2^{ème} année Cycle Supérieur (2CS)

Option : Systèmes Informatiques (SQ)

Thème : Enterprise Architecture Knowledge Graphs (EAKGs)

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Annex A: Supplementary Material

This independent annex contains supplementary material for the report. The first section provides complete SPARQL queries moved from Chapter 3; additional annex sections may be added here without changing the document's structure. The prefixes declared in the first Chapter 3 listing are assumed unless a listing explicitly introduces a new prefix.

A.1 — Exploratory Queries

```
1 SELECT DISTINCT ?component ?service ?serviceName
2 WHERE {
3   ?component rdfs:label "PaaS□Services"@en .
4   ?component (archimate:DependencyRelationship
5               |archimate:StructuralRelationship)+ ?service .
6   ?service rdf:type archimate:Element ;
7             rdfs:label ?serviceName .
8 }
```

Listing A.1: Reverse dependency chain: elements depending on a given component

```
1 # Newly introduced prefix for this listing.
2 PREFIX owl: <http://www.w3.org/2002/07/owl#>
3
4 SELECT ?s ?n ?layer
5 WHERE {
6   ?s a archimate:Element, ?layer .
7   ?layer a archimate:Layer .
8   ?s rdfs:label ?n .
9 }
```

Listing A.2: List elements with their ArchiMate layer

```
1 SELECT ?src ?target (COUNT(?rel) AS ?counter)
2 WHERE {
3   ?src rdfs:subClassOf archimate:ApplicationLayer .
4   ?target rdfs:subClassOf archimate:BusinessLayer .
5   ?a a ?src .
6   ?b a ?target .
7   ?a ?rel ?b .
8 }
9 GROUP BY ?src ?target
```

Listing A.3: Cross-layer relationship counts: Application to Business

A.2 — Gap Analysis Queries

```
1 # Newly introduced prefix for this listing.
2 PREFIX ex: <http://www.example.org/eakg#>
3
4 # Case-study IDs: baseline = ex:id-47396; target = ex:id-46339.
5
6 SELECT ?shared_element
7 WHERE {
8   ex:id-47396 archimate:Aggregation ?shared_element .
9   ex:id-46339 archimate:Aggregation ?shared_element .
10 }
```

Listing A.4: ABBs present in both architectures (Kept)

```
1 # Prefix used for the gap-analysis resources and properties.
2 PREFIX ex: <http://www.example.org/eakg#>
```

```

3 SELECT ?baseline_element
4 WHERE {
5   ex:id-47396 archimate:Aggregation ?baseline_element .
6   FILTER NOT EXISTS {
7     ex:id-46339 archimate:Aggregation ?baseline_element .
8   }
9 }
10

```

Listing A.5: ABBs present only in the baseline (candidates for elimination)

```

1 # Prefix used for the gap-analysis resources and properties.
2 PREFIX ex: <http://www.example.org/eakg#>
3
4 SELECT ?target_element
5 WHERE {
6   ex:id-46339 archimate:Aggregation ?target_element .
7   FILTER NOT EXISTS {
8     ex:id-47396 archimate:Aggregation ?target_element .
9   }
10 }

```

Listing A.6: ABBs present only in the target (new gaps)

```

1 # Prefix used for the gap-analysis resources and properties.
2 PREFIX ex: <http://www.example.org/eakg#>
3
4 # Record one reviewed new element and one eliminated element.
5
6 INSERT {
7   ?gap ex:gapNew ex:id-46919 .
8   ?gap ex:gapEliminated ex:id-46864 .
9 }
10 WHERE {
11   ?gap archimate:Association ex:id-47396 .
12   ?gap archimate:Association ex:id-46339 .
13 }

```

Listing A.7: Mark gap decisions via SPARQL Update

A.3 — Architecture Evolution Query

```

1 # Prefix used for the transformation vocabulary.
2 PREFIX ex: <http://www.example.org/eakg#>
3
4 SELECT ?transformation
5 WHERE {
6   ?transformation rdf:type ex:Transformation .
7   ?project ex:name "Back-Office_System_Integration" .
8   ?transformation ex:appliedBy ?project .
9 }

```

Listing A.8: SPARQL equivalent of a DL query for transformation lookup

A.4 — Structural and Connectivity Smells Queries

```

1 SELECT DISTINCT ?node ?label
2 WHERE {
3   ?node rdfs:label ?label .
4   ?node archimate:DependencyRelationship+ ?node .
5 }

```

Listing A.9: Detect cyclic dependencies

```

1 SELECT ?dataNode ?label
2       (COUNT(DISTINCT ?businessNode) AS ?accessCount)
3 WHERE {
4   ?dataNode a archimate:PassiveStructureElement .
5   ?dataNode rdfs:label ?label .
6   ?businessNode a archimate:BusinessLayer .
7   ?businessNode archimate:Access ?dataNode .
8 }
9 GROUP BY ?dataNode ?label
10 HAVING (COUNT(DISTINCT ?businessNode) >= 2)

```

Listing A.10: Detect shared persistency

```

1 SELECT DISTINCT ?service ?label
2 WHERE {
3   ?service a ?serviceType .
4   FILTER (?serviceType IN (archimate:ApplicationService,
5                             archimate:BusinessService))
6   ?service rdfs:label ?label .
7   ?relOut rdfs:subPropertyOf* archimate:Relationship .
8   ?service ?relOut ?dataNode .
9   ?dataNode a ?dataType .
10  ?dataType rdfs:subClassOf* archimate:PassiveStructureElement .
11  FILTER NOT EXISTS {
12    ?relOther rdfs:subPropertyOf* archimate:Relationship .
13    ?service ?relOther ?otherNode .
14    ?otherNode a ?otherType .
15    ?otherType rdfs:subClassOf* ?forbiddenClass .
16    FILTER (?forbiddenClass IN (
17      archimate:ActiveStructureElement,
18      archimate:BehaviorElement))
19  }
20 }

```

Listing A.11: Detect data-only services

A.5 — Model Quality Smells Queries

```

1 SELECT ?label (COUNT(DISTINCT ?node) AS ?instanceCount)
2 WHERE {
3   ?node rdfs:label ?label .
4   FILTER(isIRI(?node))
5 }
6 GROUP BY ?label
7 HAVING (COUNT(DISTINCT ?node) > 1)

```

Listing A.12: Detect duplicate elements by name

```

1 SELECT ?element ?comment
2       (STRLEN(STR(?comment)) AS ?length)
3 WHERE {
4   ?element rdfs:comment ?comment .
5   FILTER (STRLEN(STR(?comment)) > 400)
6 }

```

Listing A.13: Detect unnecessarily long documentation

Annex A.6 — EA Smell Tagging Queries

The following INSERT queries materialize `smells:hasSmell` triples on the elements or relations identified by the detection queries above. They use the additional prefix `smells: <http://purl.org/eakg/smells#>`.

```
1 INSERT {  
2   ?node smells:hasSmell smells:CyclicDependency .  
3 } WHERE {  
4   ?node archimate:DependencyRelationship+ ?node .  
5 }
```

Listing A.14: Tag cyclic dependency elements

```
1 INSERT { ?dataNode smells:hasSmell smells:SharedPersistency . }  
2 WHERE {  
3   {  
4     SELECT ?dataNode WHERE {  
5       ?dataNode a archimate:PassiveStructureElement .  
6       ?businessNode a archimate:BusinessLayer .  
7       ?businessNode archimate:Access ?dataNode .  
8     }  
9     GROUP BY ?dataNode  
10    HAVING (COUNT(DISTINCT ?businessNode) >= 2)  
11  }  
12 }
```

Listing A.15: Tag shared persistency data nodes

```
1 INSERT {  
2   ?relation smells:hasSmell smells:StrictLayersViolation .  
3 } WHERE {  
4   {  
5     ?nodeA a archimate:BusinessLayer .  
6     ?nodeB a archimate:TechnologyLayer .  
7     ?nodeA ?relation ?nodeB .  
8   } UNION {  
9     ?nodeA a archimate:TechnologyLayer .  
10    ?nodeB a archimate:BusinessLayer .  
11    ?nodeA ?relation ?nodeB .  
12  }  
13  ?relation rdfs:subPropertyOf* archimate:Relationship .  
14  FILTER(STRSTARTS(STR(?relation), "http://www.example.org/eakg#id"))  
15 }
```

Listing A.16: Tag strict layers violation edges

```
1 INSERT {  
2   ?service smells:hasSmell smells:ChattyService .  
3 } WHERE {  
4   {  
5     SELECT ?service WHERE {  
6       ?service a ?serviceType .  
7       FILTER (?serviceType IN (archimate:ApplicationService,  
8                                archimate:BusinessService))  
9       { ?service ?relation ?connectedElement . }  
10      UNION  
11      { ?connectedElement ?relation ?service . }  
12      ?relation rdfs:subPropertyOf* archimate:Relationship .  
13      FILTER(STRSTARTS(STR(?relation),  
14                    "http://www.example.org/eakg#id"))  
15    }  
16    GROUP BY ?service  
17    HAVING (COUNT(DISTINCT ?connectedElement) >= 10)  
18  }  
19 }
```

Listing A.17: Tag chatty service nodes

```

1 INSERT {
2   ?service smells:hasSmell smells:DataService .
3 } WHERE {
4   ?service a ?serviceType .
5   FILTER (?serviceType IN (archimate:ApplicationService,
6                             archimate:BusinessService))
7   ?relOut rdfs:subPropertyOf* archimate:Relationship .
8   ?service ?relOut ?dataNode .
9   ?dataNode a ?dataType .
10  ?dataType rdfs:subClassOf* archimate:PassiveStructureElement .
11  FILTER NOT EXISTS {
12    ?relOther rdfs:subPropertyOf* archimate:Relationship .
13    ?service ?relOther ?otherNode .
14    ?otherNode a ?otherType .
15    ?otherType rdfs:subClassOf* ?forbiddenClass .
16    FILTER (?forbiddenClass IN (
17      archimate:ActiveStructureElement,
18      archimate:BehaviorElement))
19  }
20 }

```

Listing A.18: Tag data-only service nodes

```

1 INSERT { ?node smells:hasSmell smells:Duplication . }
2 WHERE {
3   ?node rdfs:label ?label .
4   FILTER(isIRI(?node))
5   {
6     SELECT ?label WHERE {
7       ?innerNode rdfs:label ?label .
8       FILTER(isIRI(?innerNode))
9     }
10    GROUP BY ?label
11    HAVING (COUNT(DISTINCT ?innerNode) > 1)
12  }
13 }

```

Listing A.19: Tag duplicate elements

```

1 INSERT { ?resource smells:hasSmell smells:UnnecessaryDocumentation . }
2 WHERE {
3   ?resource rdfs:comment ?comment .
4   FILTER (!STRSTARTS(STR(?resource), "http://purl.org/eakg/archimate"))
5   FILTER (!STRSTARTS(STR(?resource), "http://purl.org/eakg/smells"))
6   FILTER (STRLEN(STR(?comment)) > 400)
7 }

```

Listing A.20: Tag unnecessarily long documentation

Annex A.7 — Smell Tag Utility Queries

Replace `smells:CyclicDependency` with any smell URI to parameterize these utility queries.

```
1  DELETE WHERE {  
2    ?element smells:hasSmell smells:CyclicDependency .  
3  }
```

Listing A.21: Reset (delete) tags for a specific smell

Annex B: Implementation Detail

This annex records the implementation details omitted from Chapter 2 for readability.

B.1 — Utility Functions

- `id_to_iri(identifier)` converts element and relationship identifiers to the `ex:` namespace.
- `property_name_to_iri(key_name)` sanitizes property keys and prevents digit-leading local names.
- `resolve_archimate_class(xsi_type_value)` maps XML `xsi:type` values to `archimate:` URIs.
- `emit_lang_literals(...)` emits language-tagged literals for names and documentation, falling back to `xsd:string` when no language is supplied.

B.2 — Translator Passes

1. **Pass 1:** read each property definition, resolve its datatype, create its sanitized predicate IRI, emit `owl:DatatypeProperty`, `rdfs:subPropertyOf archimate:Property`, and `rdfs:range` triples, then store the predicate/datatype pair by definition ID.
2. **Pass 2:** convert each element ID to an IRI, resolve its `ArchiMate` class, declare the named individual and its type, emit multilingual labels/comments, and resolve property values through the Pass 1 dictionary.
3. **Pass 3:** convert each relationship ID into an object-property predicate, declare its `ArchiMate` sub-property, attach documentation, emit the source-to-target edge, and resolve relationship property values through the same dictionary.

Annex C — Detailed AI Evaluation Results

* indicates that hallucination was detected in the answer. – indicates a failed run.

Table 1: Question-level final scores for the four test conditions

Q	XML-R1	XML-R2	JSON-R1	JSON-R2	TTL-R1	TTL-R2	DB-R1	DB-R2
1	0.18*	1.00	1.00	0.65	0.25*	0.83	0.15	1.00
2	1.00	1.00	1.00	1.00	0.65	0.65	1.00	1.00
3	1.00	0.83	1.00	1.00	0.82	0.83	1.00	1.00
4	0.65	0.25*	0.25*	0.48	0.25*	1.00	1.00	1.00
5	0.25*	0.65	0.85	0.18*	0.25*	1.00	1.00	1.00
6	0.25*	0.83	1.00	1.00	0.65	1.00	1.00	1.00
7	0.00*	1.00	1.00	1.00	0.65	1.00	1.00	1.00
8	0.25*	0.25*	0.00	0.83	0.65	0.00	1.00	1.00
9	1.00	1.00	1.00	1.00	1.00	1.00	0.18*	0.85
10	1.00	1.00	1.00	0.83	0.25*	0.25*	0.70	0.85
11	1.00	1.00	0.00	1.00	0.25*	0.83	0.70	0.85
12	1.00	1.00	0.25*	1.00	1.00	1.00	0.85	0.85
13	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.85
14	0.00	1.00	1.00	1.00	0.82	1.00	0.85	0.85
15	1.00	1.00	1.00	0.83	0.25*	0.25*	0.70	0.18
16	1.00	1.00	0.83	1.00	0.00	1.00	1.00	1.00
17	0.00	1.00	0.00	0.00	0.65	0.00*	1.00	0.85
18	1.00	1.00	0.83	1.00	0.82	0.65	–	0.85
19	0.00	1.00	1.00	1.00	0.30	1.00	1.00	1.00
20	1.00	1.00	1.00	1.00	0.30	1.00	1.00	1.00
21	1.00	1.00	1.00	1.00	0.82	0.25*	1.00	0.85
22	1.00	1.00	1.00	1.00	1.00	0.00	1.00	0.85
23	1.00	1.00	0.00	1.00	0.25*	1.00	1.00	1.00
24	0.65	0.65	0.25*	1.00	0.25*	1.00	1.00	0.85
25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
26	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00